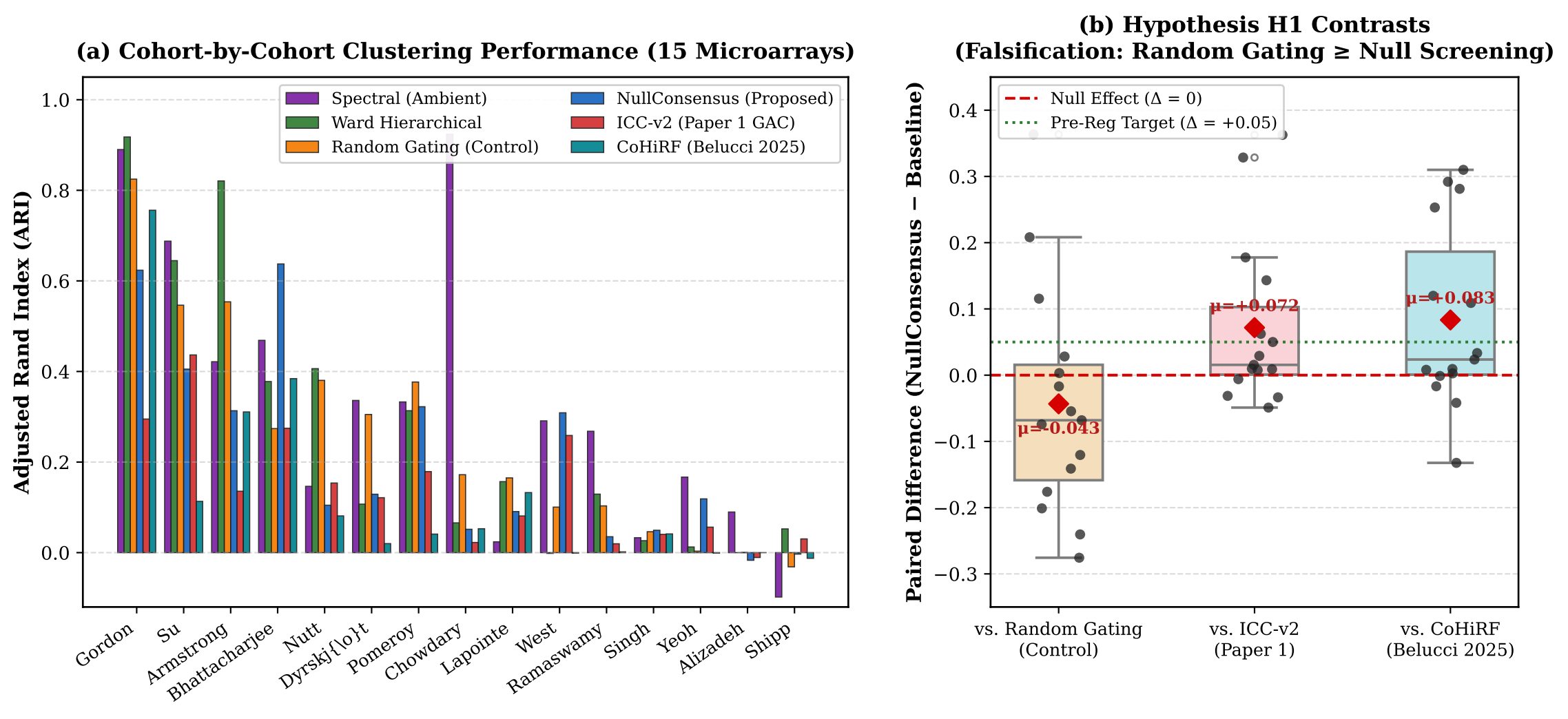
(a) Cohort-by-Cohort Clustering Performance (15 Microarrays)
Adjusted Rand Index (ARI)
Spectral (Ambient)
Ward Hierarchical
Random Gating (Control)
NullConsensus (Proposed)
ICC-v2 (Paper 1 GAC)
CoHiRF (Belucci 2025)
Gordon
Su
Armstrong
Bhattacharjee
Nutt
Dyrskj{\o}t
Pomeroy
Chowdary
Lapointe
West
Ramaswamy
Singh
Yeoh
Alizadeh
Shipp
(b) Hypothesis H1 Contrasts
(Falsification: Random Gating ≥ Null Screening)
Paired Difference (NullConsensus − Baseline)
Null Effect (Δ = 0)
Pre-Reg Target (Δ = +0.05)
μ=-0.043
μ=+0.072
μ=+0.083
vs. Random Gating
(Control)
vs. ICC-v2
(Paper 1)
vs. CoHiRF
(Belucci 2025)